



ENTERPRISE SOC

THREAT DETECTION, MONITORING & INCIDENT RESPONSE LAB



SPLUNK • WAZUH • SYSMON • WIRESHARK • KALI LINUX



BLUE TEAM ANALYST

CYBER ATTACK KILL CHAIN



THREAT DETECTION
Real-time Monitoring
& Alerting



SIEM & LOG ANALYSIS
Splunk Dashboard
& Correlation



ENDPOINT MONITORING
Sysmon & Windows
Telemetry



NETWORK TRAFFIC ANALYSIS
Wireshark Packet
Inspection



WAZUH SECURITY
Log Analysis &
Threat Intelligence



INCIDENT RESPONSE
Triage, Investigation
& Containment



PREPARED BY:
PARAG PATEL
CYBERSECURITY PROJECT PORTFOLIO



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1. Project Overview

Introduction

Modern organizations generate massive volumes of security events from endpoints, applications, servers, and network devices every second. Manual monitoring of such large-scale security telemetry is inefficient, time-consuming, and increases the risk of delayed threat detection and incident response.

Security Operations Centers (SOC) rely on centralized monitoring platforms, endpoint detection solutions, and network visibility tools to identify suspicious behavior, investigate security incidents, and improve organizational cyber resilience.

The purpose of this project is to design and implement an advanced enterprise-style SOC threat detection, monitoring, and incident response lab using multiple security technologies including Splunk Enterprise, Wazuh, Sysmon, Wireshark, Windows endpoint monitoring, and Kali Linux attack simulation.

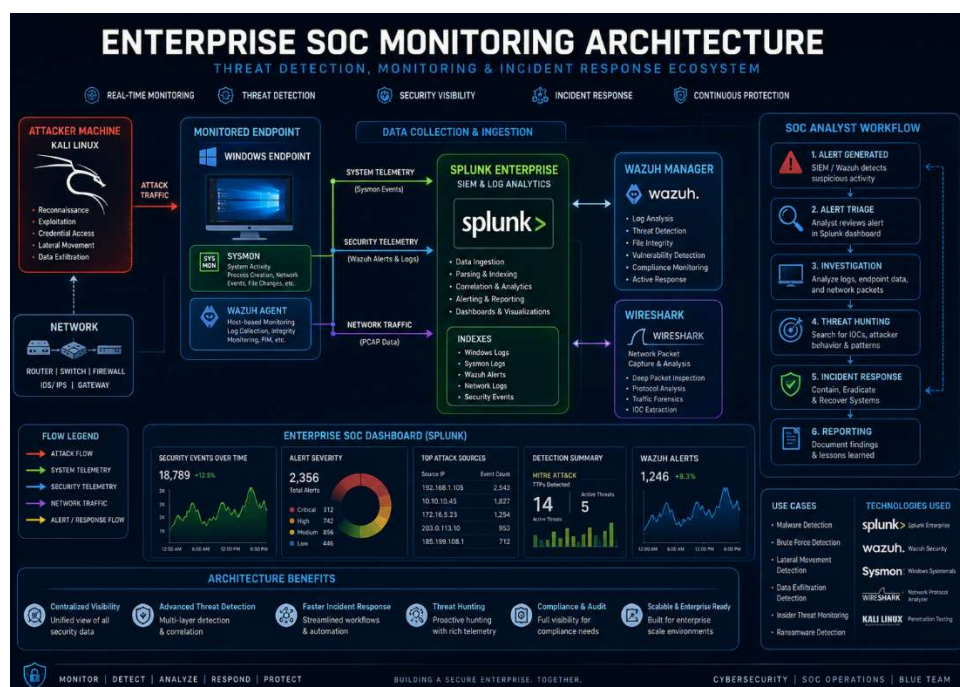


Figure 1: Enterprise SOC Monitoring Architecture Overview

This practical lab simulates realistic enterprise cyber defense workflows by integrating endpoint telemetry collection, centralized SIEM monitoring, attack detection engineering, alert correlation, packet-level network analysis, and incident investigation workflows.

The environment includes a monitored Windows host machine, Kali Linux attacker system, Splunk Enterprise SIEM platform, Wazuh security monitoring, Sysmon endpoint telemetry, and Wireshark packet inspection for multi-layer threat visibility.

This implementation focuses on practical real-world threat scenarios including reconnaissance activity, brute force authentication abuse, suspicious PowerShell execution, unauthorized process monitoring, and network traffic investigation.

2. Project Objectives

The primary objective of this project is to design and implement an enterprise-grade Security Operations Center (SOC) monitoring and incident response lab using practical cybersecurity tools and enterprise security workflows.

This project focuses on building hands-on defensive cybersecurity capabilities by integrating endpoint monitoring, SIEM-based log analysis, threat detection engineering, network traffic monitoring, and alert investigation.

The specific objectives of this project are:

- To deploy and configure Splunk Enterprise as a centralized SIEM platform for security log monitoring and analysis.
- To integrate Wazuh for endpoint threat detection, alert monitoring, and host-based security visibility.
- To configure Sysmon for advanced Windows telemetry collection including process execution, network connections, and endpoint behavior monitoring.
- To utilize Wireshark for packet-level network traffic inspection and suspicious communication analysis.
- To simulate realistic cyber-attack scenarios including reconnaissance scanning, brute force authentication abuse, suspicious PowerShell activity, and unauthorized process execution.
- To develop custom detection logic using Splunk Search Processing Language (SPL) queries.
- To investigate alerts and suspicious activity using practical SOC analyst workflows.
- To improve practical blue team skills related to log analysis, threat detection, security monitoring, and incident response.
- To align observed attack behavior with the MITRE ATT&CK framework for threat intelligence mapping.
- To build a recruiter-focused enterprise cybersecurity project demonstrating practical SOC analyst capabilities.

3. Architecture Design

SOC Lab Architecture

The architecture of this enterprise SOC lab was designed to simulate a practical multi-layer cybersecurity monitoring environment used in real-world defensive security operations.

The monitored Windows host machine acts as the primary endpoint generating security telemetry, process activity, authentication logs, PowerShell execution events, and system behavior data.

Splunk Enterprise serves as the centralized Security Information and Event Management (SIEM) platform responsible for collecting, indexing, searching, and correlating security logs for threat detection and incident investigation.

Wazuh provides host-based intrusion detection, alert generation, endpoint monitoring, and security event visibility to enhance detection coverage.

Sysmon enables advanced Windows event logging by capturing process creation, network connections, command-line activity, and suspicious endpoint behavior.

Wireshark is integrated for packet-level traffic inspection and network behavior analysis during attack simulation and investigation workflows.

The Kali Linux attacker machine simulates realistic offensive activity including reconnaissance scanning, brute force authentication attempts, and suspicious security testing actions.

This architecture provides layered visibility across endpoint telemetry, SIEM monitoring, host-based alerts, and network traffic inspection for practical SOC operations.

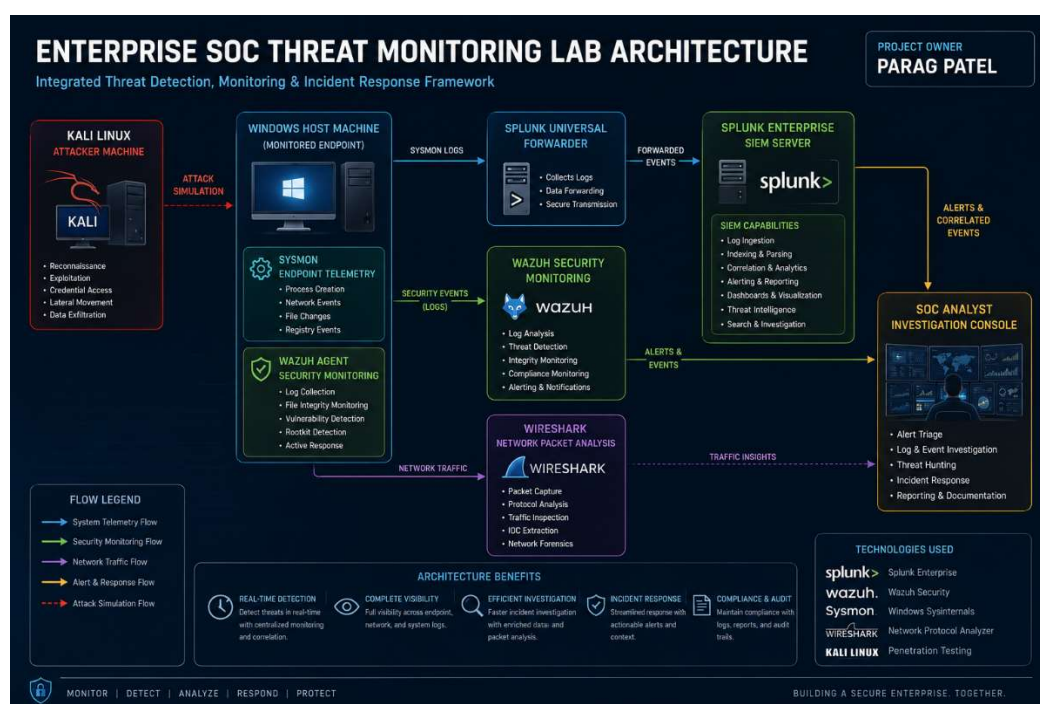


Figure 2: Enterprise SOC Lab Architecture Design

4. Tools and Technologies Used

Tool / Technology	Purpose
Splunk Enterprise	Centralized SIEM log monitoring, analysis, and security investigation
Wazuh	Host-based threat detection, alert monitoring, and endpoint visibility
Sysmon	Advanced Windows endpoint telemetry collection and process monitoring
Splunk Universal Forwarder	Secure transmission of Windows logs to Splunk Enterprise
Wireshark	Packet-level network traffic analysis and suspicious communication monitoring
Kali Linux	Cyber-attack simulation and security testing platform
Windows Host Machine	Monitored enterprise endpoint generating security telemetry
Sysinternals Tools	Administrative and suspicious process execution simulation
Nmap / PowerShell	Reconnaissance scanning and attack activity simulation

5. System Requirements

This enterprise SOC monitoring lab requires adequate hardware resources to support SIEM operations, endpoint monitoring, network traffic analysis, and attack simulation within a controlled cybersecurity environment.

The project architecture utilizes the host Windows machine as the monitored enterprise endpoint, while Kali Linux operates as the attacker virtual machine for security testing activities.

Component	Recommended Specification
Processor	Intel Core i5 / i7 or AMD Ryzen equivalent
RAM	Minimum 16 GB (32 GB preferred)
Storage	150 GB+ free SSD space
Operating System	Windows 10 / Windows 11 Host Machine
Virtualization	Intel VT-x / AMD-V Enabled
Network	Stable virtual networking connectivity

Security Component	Role
Windows Host Machine	Monitored enterprise endpoint
Kali Linux VM	Attack simulation environment
Splunk Enterprise	SIEM monitoring platform
Wazuh	Endpoint detection and alerting
Sysmon	Advanced Windows telemetry monitoring
Wireshark	Network packet analysis

6. Environment Setup and Configuration

6.1 Virtual Lab Architecture Setup

To simulate an enterprise-style Security Operations Center (SOC) monitoring environment, a controlled virtual lab infrastructure was established using the host Windows machine and dedicated security monitoring tools.

The Windows host system was configured as the primary monitored enterprise endpoint responsible for generating authentication logs, process activity telemetry, PowerShell events, and endpoint security data.

A Kali Linux virtual machine was deployed as the attacker platform for simulating realistic cyber attack scenarios such as reconnaissance scanning, authentication abuse, and suspicious network activity.

Splunk Enterprise was configured as the centralized SIEM platform for log collection, search analysis, and security event correlation.

Wazuh was integrated to enhance endpoint monitoring, alert generation, and host-based intrusion detection capabilities.

Wireshark was prepared for packet-level network inspection and attack traffic analysis.

This architecture provides a practical multi-layer enterprise monitoring environment aligned with real-world SOC analyst workflows.

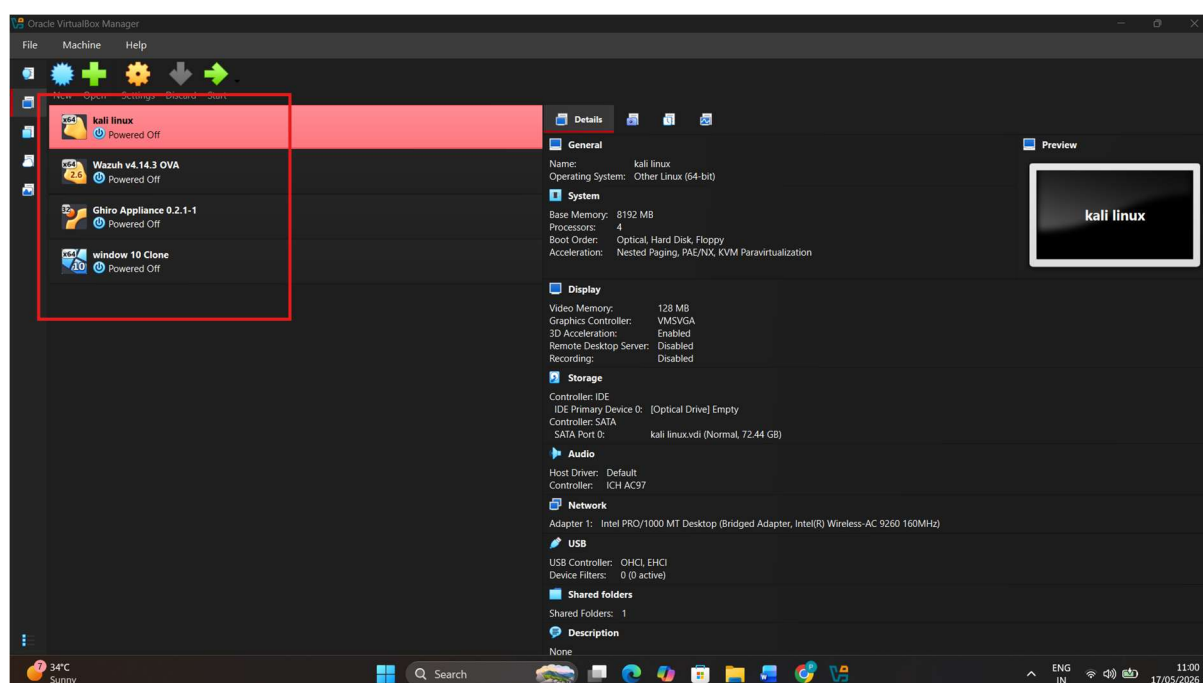


Figure 3: Virtual SOC Lab Environment with Kali Linux Attacker Machine

6.2 Windows Endpoint Monitoring Configuration

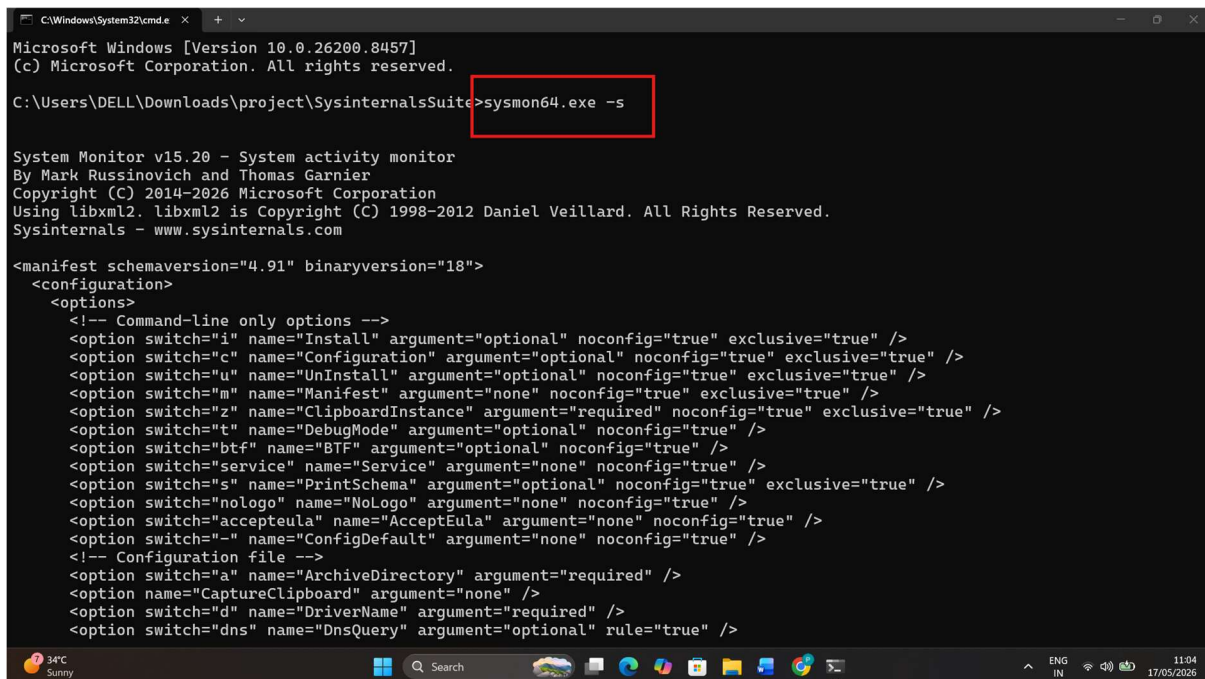
The Windows host machine was configured as the primary monitored enterprise endpoint within the SOC monitoring environment.

This endpoint serves as the main source of security telemetry including Windows Security Event Logs, authentication records, PowerShell activity, process execution events, and endpoint behavioral data.

To strengthen endpoint visibility, Sysmon was deployed to generate enhanced telemetry such as process creation events, command-line execution activity, network connections, and suspicious endpoint behavior.

The Windows endpoint was also prepared for centralized monitoring through Splunk Universal Forwarder integration and Wazuh endpoint monitoring.

This configuration establishes practical enterprise endpoint visibility required for threat detection, security monitoring, and SOC investigation workflows.



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL\Downloads\project\SysinternalsSuite>sysmon64.exe -s

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

<manifest schemaversion="4.91" binaryversion="18">
  <configuration>
    <options>
      <!-- Command-line only options -->
      <option switch="i" name="Install" argument="optional" noconfig="true" exclusive="true" />
      <option switch="c" name="Configuration" argument="optional" noconfig="true" exclusive="true" />
      <option switch="u" name="UnInstall" argument="optional" noconfig="true" exclusive="true" />
      <option switch="m" name="Manifest" argument="none" noconfig="true" exclusive="true" />
      <option switch="z" name="ClipboardInstance" argument="required" noconfig="true" exclusive="true" />
      <option switch="t" name="DebugMode" argument="optional" noconfig="true" />
      <option switch="btf" name="BTF" argument="optional" noconfig="true" />
      <option switch="service" name="Service" argument="none" noconfig="true" />
      <option switch="s" name="PrintSchema" argument="optional" noconfig="true" exclusive="true" />
      <option switch="nologo" name="NoLogo" argument="none" noconfig="true" />
      <option switch="accepteula" name="AcceptEula" argument="none" noconfig="true" />
      <option switch="-" name="ConfigDefault" argument="none" noconfig="true" />
      <!-- Configuration file -->
      <option switch="a" name="ArchiveDirectory" argument="required" />
      <option name="CaptureClipboard" argument="none" />
      <option switch="d" name="DriverName" argument="required" />
      <option switch="dns" name="DnsQuery" argument="optional" rule="true" />
    </options>
  </configuration>
</manifest>
```

Figure 4: Windows Host Endpoint Monitoring Configuration using Sysmon

6.3 Splunk Enterprise SIEM Configuration

Splunk Enterprise was deployed as the centralized Security Information and Event Management (SIEM) platform within the enterprise SOC monitoring environment.

The platform was configured to collect, index, search, and analyze security telemetry generated by the monitored Windows host endpoint.

Splunk enables centralized visibility into authentication events, process execution activity, Sysmon telemetry, suspicious endpoint behavior, and security investigation workflows.

Custom search queries were used to validate event ingestion, identify suspicious activities, and support practical SOC detection workflows.

The deployment of Splunk Enterprise forms the core monitoring and threat analysis layer of the SOC lab architecture.

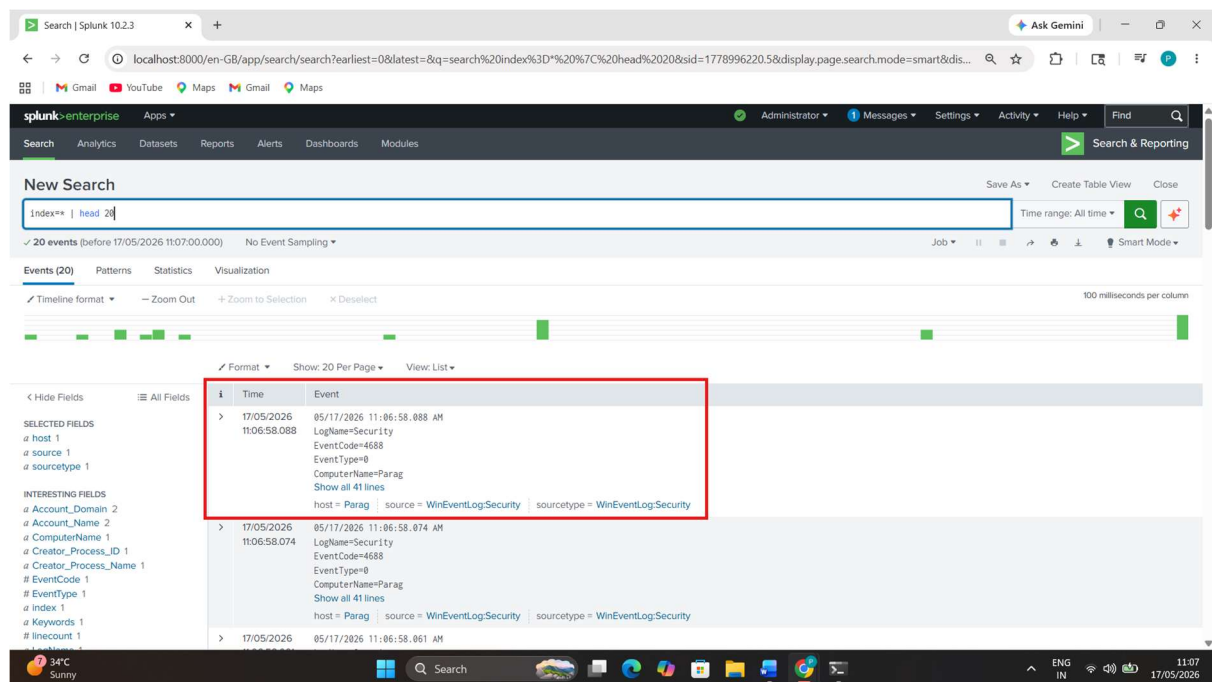


Figure 5: Splunk Enterprise SIEM Security Monitoring Dashboard

6.4 Wazuh Deployment and Alert Monitoring

Wazuh was integrated into the enterprise SOC monitoring environment to enhance endpoint security visibility, host-based intrusion detection, and alert monitoring capabilities.

The Wazuh platform provides real-time monitoring of endpoint activity, security alerts, system behavior analysis, file integrity monitoring, and suspicious activity detection.

By integrating Wazuh alongside Splunk Enterprise, the SOC lab achieves layered detection capability across SIEM monitoring and endpoint-focused alert analysis.

Wazuh alerts assist in identifying suspicious endpoint behavior, security anomalies, authentication events, and threat indicators that support practical SOC investigation workflows.

This integration strengthens the enterprise monitoring architecture by improving threat visibility and alert correlation.

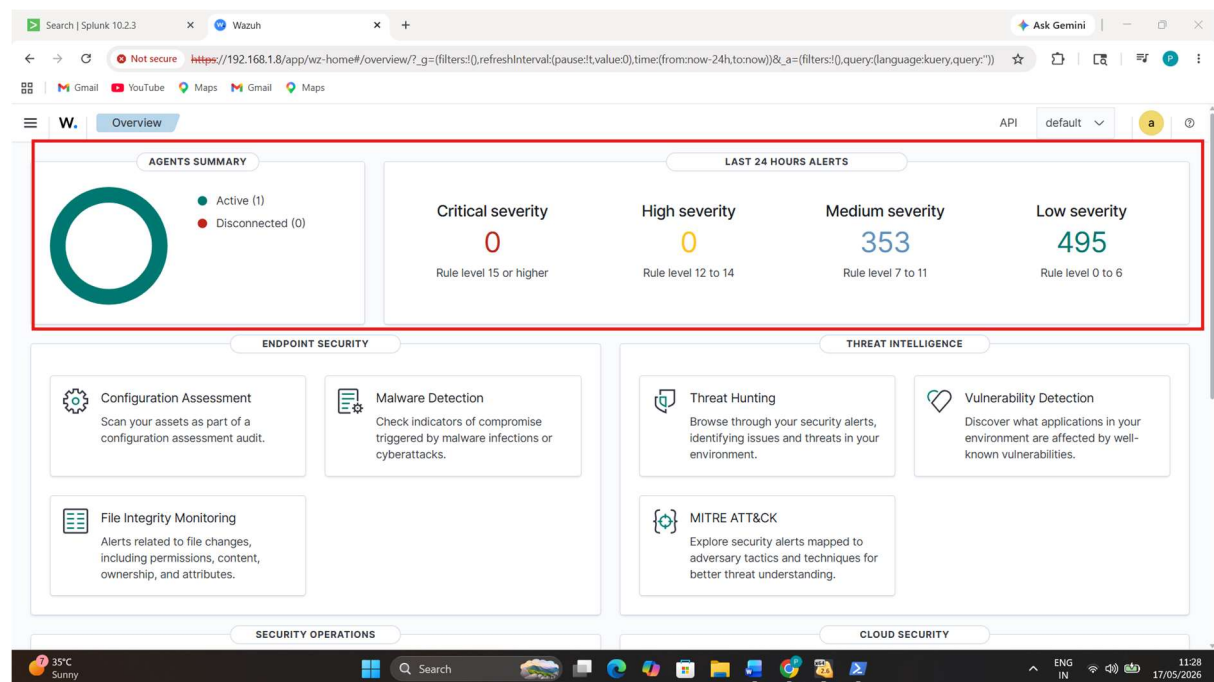


Figure 6: Wazuh Security Monitoring and Alert Management Dashboard

6.5 Sysmon Deployment and Endpoint Telemetry

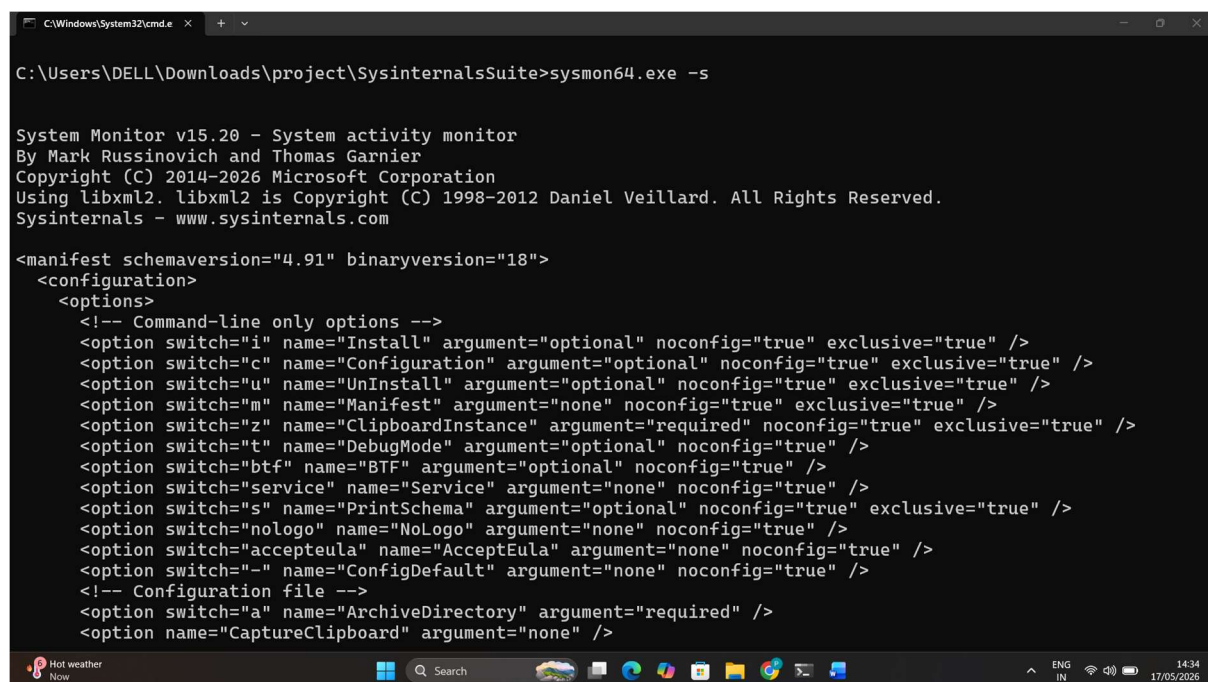
Sysmon was deployed on the Windows host machine to enhance endpoint visibility by generating advanced security telemetry beyond standard Windows event logging.

Sysmon captures detailed endpoint activity including process creation events, command-line execution, network connections, driver loading, and suspicious system behavior.

This telemetry significantly improves threat detection capabilities by enabling visibility into endpoint behavior commonly associated with attacker activity.

The generated Sysmon logs are forwarded to Splunk Enterprise for centralized monitoring and security investigation workflows.

Sysmon integration provides practical enterprise-grade endpoint telemetry essential for SOC threat detection and incident analysis.



```
C:\Windows\System32\cmd.exe
C:\Users\DELL\Downloads\project\SysinternalsSuite>sysmon64.exe -s

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

<manifest schemaversion="4.91" binaryversion="18">
  <configuration>
    <options>
      <!-- Command-line only options -->
      <option switch="i" name="Install" argument="optional" noconfig="true" exclusive="true" />
      <option switch="c" name="Configuration" argument="optional" noconfig="true" exclusive="true" />
      <option switch="u" name="UnInstall" argument="optional" noconfig="true" exclusive="true" />
      <option switch="m" name="Manifest" argument="none" noconfig="true" exclusive="true" />
      <option switch="z" name="ClipboardInstance" argument="required" noconfig="true" exclusive="true" />
      <option switch="t" name="DebugMode" argument="optional" noconfig="true" />
      <option switch="btf" name="BTF" argument="optional" noconfig="true" />
      <option switch="service" name="Service" argument="none" noconfig="true" />
      <option switch="nologo" name="NoLogo" argument="none" noconfig="true" />
      <option switch="accepteula" name="AcceptEula" argument="none" noconfig="true" />
      <option switch="-" name="ConfigDefault" argument="none" noconfig="true" />
      <!-- Configuration file -->
      <option switch="a" name="ArchiveDirectory" argument="required" />
      <option name="CaptureClipboard" argument="none" />
    </options>
  </configuration>
</manifest>
```

Figure 7: Sysmon Endpoint Telemetry Configuration on Windows Host

6.6 Splunk Universal Forwarder Configuration

Splunk Universal Forwarder was configured on the Windows host machine to securely transmit endpoint logs and security telemetry to the centralized Splunk Enterprise SIEM platform.

The forwarder enables real-time delivery of Windows Security Event Logs, Sysmon telemetry, process activity records, and authentication events for centralized monitoring and investigation.

This integration ensures continuous log collection from the monitored endpoint, allowing Splunk to perform indexing, searching, threat detection, and event correlation.

The successful deployment of the Splunk Universal Forwarder establishes reliable communication between the Windows host machine and the SIEM infrastructure.

This component is critical for maintaining real-time enterprise security monitoring workflows.

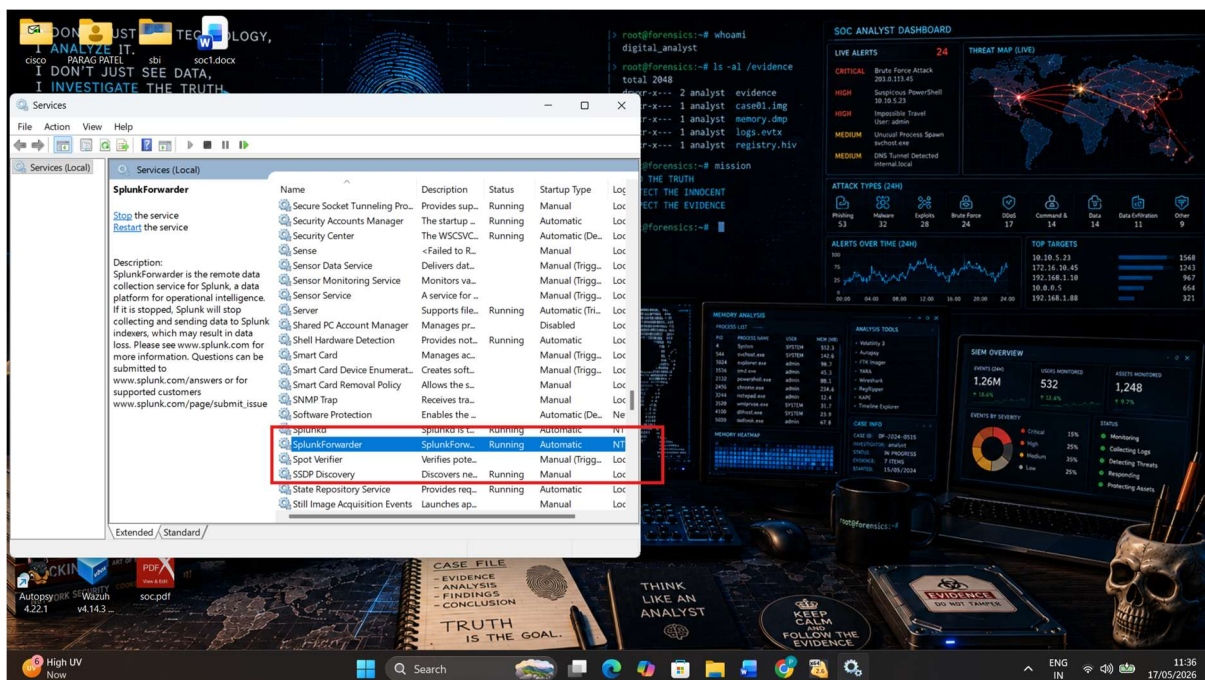


Figure 8: Splunk Universal Forwarder Service Running on Windows Host

6.7 Wireshark Network Monitoring Setup

Wireshark was integrated into the enterprise SOC monitoring environment to provide packet-level network visibility during attack simulation and security investigation activities.

The tool enables detailed inspection of network communication, protocol behavior, connection attempts, and suspicious traffic patterns generated during simulated attack scenarios.

Wireshark supports practical network forensics workflows by allowing analysis of inbound and outbound traffic between the monitored Windows host machine and the Kali Linux attacker environment.

This layer improves SOC visibility beyond endpoint telemetry by enabling packet inspection and communication-level threat analysis.

The integration of Wireshark strengthens practical enterprise incident investigation and network monitoring capabilities.

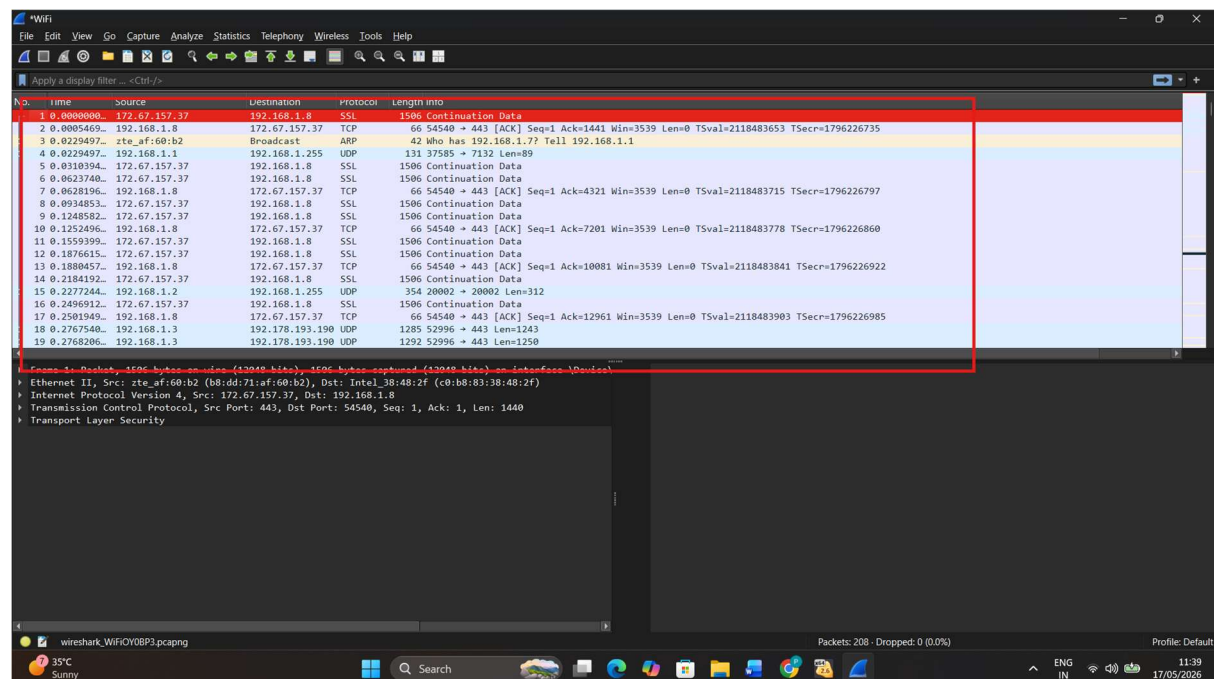


Figure 9: Wireshark Network Traffic Monitoring on Windows Host Machine

6.8 Log Collection Validation

Log collection validation was performed to confirm successful ingestion of endpoint telemetry, Windows security events, Sysmon logs, and forwarded security data into the centralized Splunk Enterprise SIEM platform.

Test search queries were executed to verify that the monitored Windows host machine was successfully transmitting real-time security events to Splunk.

The validation process confirmed successful indexing, search visibility, and centralized monitoring of security telemetry required for practical threat detection workflows.

This verification ensured that the SOC monitoring environment was fully operational before initiating attack simulation and detection testing activities.

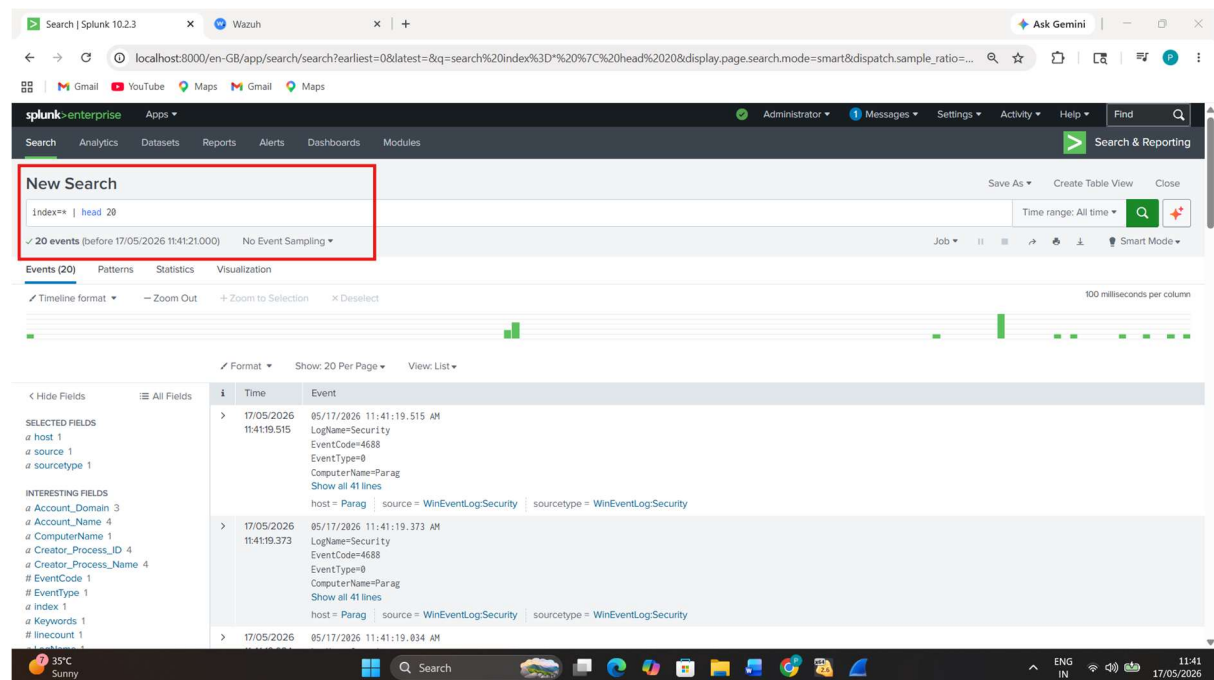


Figure 10: Successful Log Collection Validation in Splunk Enterprise

7. Threat Simulation, Detection, and Incident Response

7.1 Attack Simulation Overview

To evaluate the effectiveness of the enterprise SOC monitoring environment, practical cyber-attack simulation scenarios were performed in a controlled lab environment.

The objective of these simulations was to generate realistic security telemetry, validate detection visibility, and test incident investigation workflows across multiple monitoring layers.

Attack simulation activities included reconnaissance scanning, brute force authentication abuse, suspicious PowerShell execution, unauthorized process monitoring, and network traffic observation.

The monitored Windows host machine generated endpoint telemetry, authentication events, Sysmon process logs, and Windows security records, while Kali Linux served as the attacker platform for simulated offensive activity.

Splunk Enterprise, Wazuh, Sysmon, and Wireshark collectively provided layered visibility for practical threat detection and SOC investigation workflows.

This section demonstrates practical blue team monitoring capabilities through controlled enterprise-style threat simulation exercises.

7.2 Reconnaissance Detection (Nmap Scan)

Reconnaissance activity is commonly one of the earliest phases of cyber-attacks, where attackers attempt to discover live systems, open ports, and accessible services before further exploitation.

To simulate reconnaissance behavior, the Kali Linux attacker machine executed Nmap scanning activity against the monitored Windows host machine within the controlled SOC lab environment.

This activity generated network communication, endpoint telemetry, and security-relevant events observable across the monitoring stack.

The reconnaissance activity generated observable network traffic and security telemetry for practical investigation.

This simulation demonstrates early-stage attack detection capability within the enterprise SOC monitoring environment.

```
kali linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

root@kali:/home/kali

root@kali:~# nmap -sS 192.168.1.3
Starting Nmap 7.90 (https://nmap.org) at 2026-05-17 11:48 +0530
Nmap scan report for 192.168.1.3 (192.168.1.3)
Host is up (0.00049s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server
5357/tcp   open  wsdapi
MAC Address: C0:B8:83:38:48:2F (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 5.40 seconds

root@kali:~# nmap -A 192.168.1.3
Starting Nmap 7.90 (https://nmap.org) at 2026-05-17 11:49 +0530
Nmap scan report for 192.168.1.3 (192.168.1.3)
Host is up (0.0010s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
139/tcp    open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3389/tcp   open  ms-wbt-server
rdp-ntlm-info:
|_ Target_Name: PARAG
|_ NetBIOS_Domain_Name: PARAG
|_ NetBIOS_Computer_Name: PARAG
|_ DNS_Domain_Name: Parag
|_ DNS_Computer_Name: Parag
|_ Product_Version: 10.0.26100
|_ System_Time: 2026-05-17T06:19:35+00:00
|_ ssl-date: TLS randomness does not represent time
|_ ssl-cert: Subject: commonName=Parag
|_ Not valid before: 2026-05-12T13:05:17
```

Figure 11: Nmap Reconnaissance Scan from Kali Linux Attacker Machine

7.3 Brute Force Authentication Abuse Detection

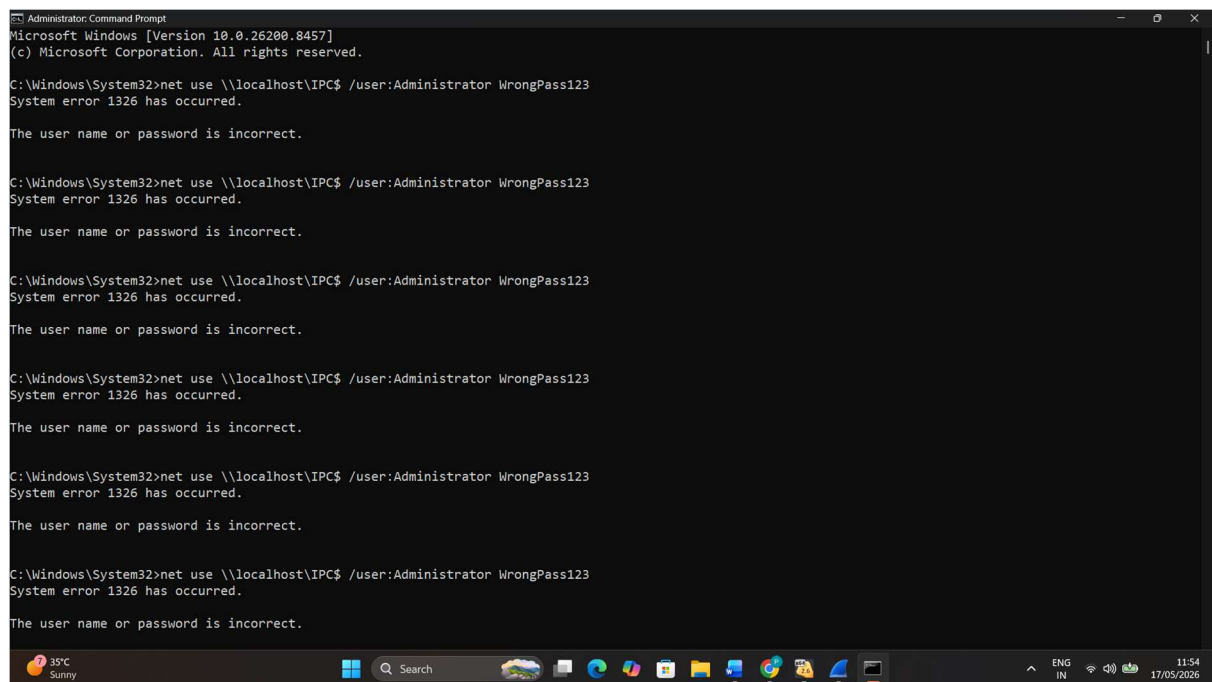
Brute force authentication abuse is a common cyber-attack technique in which attackers repeatedly attempt invalid login requests to gain unauthorized access to target systems.

To simulate authentication abuse activity, repeated failed login attempts were intentionally generated against the monitored Windows host machine within the controlled SOC lab environment.

This simulation generated Windows authentication failure events, endpoint telemetry, and centralized SIEM logs that support practical detection engineering and incident investigation.

Splunk Enterprise and Wazuh monitoring provided visibility into authentication abuse behavior through failed logon event analysis and alert monitoring workflows.

This simulation demonstrates practical enterprise capability to detect suspicious authentication activity and brute force attack indicators.



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>net use \\localhost\IPC$ /user:Administrator WrongPass123
System error 1326 has occurred.

The user name or password is incorrect.

C:\Windows\System32>net use \\localhost\IPC$ /user:Administrator WrongPass123
System error 1326 has occurred.

The user name or password is incorrect.

C:\Windows\System32>net use \\localhost\IPC$ /user:Administrator WrongPass123
System error 1326 has occurred.

The user name or password is incorrect.

C:\Windows\System32>net use \\localhost\IPC$ /user:Administrator WrongPass123
System error 1326 has occurred.

The user name or password is incorrect.

C:\Windows\System32>net use \\localhost\IPC$ /user:Administrator WrongPass123
System error 1326 has occurred.

The user name or password is incorrect.

C:\Windows\System32>net use \\localhost\IPC$ /user:Administrator WrongPass123
System error 1326 has occurred.

The user name or password is incorrect.
```

Figure 12: Simulated Brute Force Authentication Abuse Activity on Windows Host Machine

7.4 Suspicious PowerShell Activity Detection

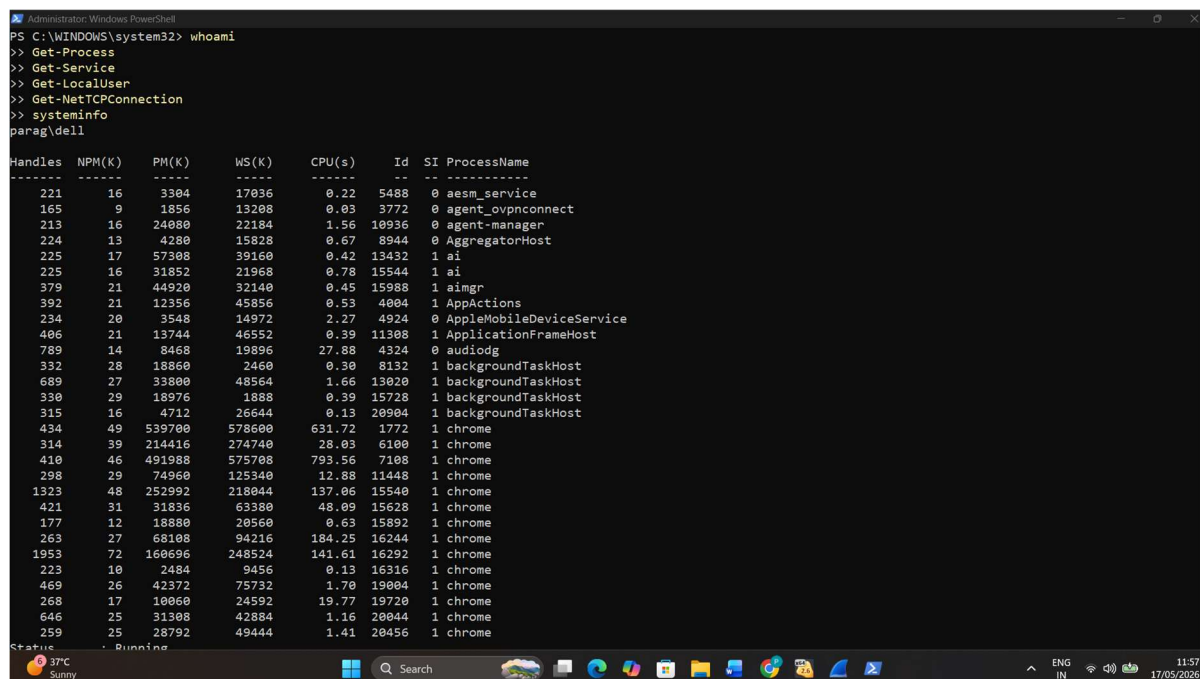
PowerShell is a legitimate administrative tool commonly used within enterprise Windows environments. However, attackers frequently abuse PowerShell for malicious execution, reconnaissance, credential access, persistence, and post-exploitation activities.

To simulate suspicious PowerShell behavior, controlled administrative commands were executed on the monitored Windows host machine to generate endpoint telemetry and security events.

The activity produced process creation logs, PowerShell execution records, and Sysmon telemetry observable through centralized SOC monitoring systems.

This simulation demonstrates practical detection engineering capability for identifying suspicious command execution behavior commonly associated with attacker techniques.

PowerShell monitoring is a critical enterprise SOC use case due to the frequent abuse of script-based administrative tools by threat actors.



```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> whoami
>> Get-Process
>> Get-Service
>> Get-LocalUser
>> Get-NetTCPConnection
>> systeminfo
parag\de11
```

Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
221	16	3304	17036	0.22	5488	0	aesm_service
165	9	1856	13208	0.03	3772	0	agent_ovpnconnect
213	16	24080	22184	1.56	10936	0	agent-manager
224	13	4280	15828	0.67	8944	0	AggregatorHost
225	17	57308	39160	0.42	13432	1	ai
225	16	31852	21968	0.78	15544	1	ai
379	21	44920	32140	0.45	15988	1	aimgr
392	21	12356	45856	0.53	4804	1	AppActions
234	20	3548	14972	2.27	4924	0	AppleMobileDeviceService
406	21	13744	46552	0.39	11308	1	ApplicationFrameHost
789	14	8468	19896	27.88	4324	0	audiodg
332	28	18860	2460	0.30	8132	1	backgroundTaskHost
689	27	33800	48564	1.66	13020	1	backgroundTaskHost
330	29	18976	1888	0.39	15728	1	backgroundTaskHost
315	16	4712	26644	0.13	20904	1	backgroundTaskHost
434	49	539700	578600	631.72	1772	1	chrome
314	39	214416	274740	28.03	6100	1	chrome
410	46	491988	575708	793.56	7108	1	chrome
298	29	74960	125340	12.88	11448	1	chrome
1323	48	252992	218044	137.06	15540	1	chrome
421	31	31836	63380	48.09	15628	1	chrome
177	12	18880	20560	0.63	15892	1	chrome
263	27	68108	94216	184.25	16244	1	chrome
1953	72	160696	248524	141.61	16292	1	chrome
223	10	2484	9456	0.13	16316	1	chrome
469	26	42372	75732	1.70	19004	1	chrome
268	17	10060	24592	19.77	19720	1	chrome
646	25	31308	42884	1.16	20044	1	chrome
259	25	28792	49444	1.41	20456	1	chrome

Figure 13: Suspicious PowerShell Activity Simulation on Windows Host Machine

7.5 Unauthorized Process Execution Monitoring

Unauthorized or suspicious process execution is a common indicator of malicious activity within enterprise environments, including attacker tool execution, post-exploitation actions, and unauthorized administrative activity.

To simulate suspicious endpoint behavior, process execution activity was intentionally generated on the monitored Windows host machine using administrative utilities and command execution tools.

This activity produced Sysmon process creation telemetry, Windows event logs, and endpoint behavioral data that support practical SOC detection workflows.

Monitoring process execution enables security analysts to identify abnormal command execution patterns, suspicious parent-child process relationships, and unauthorized endpoint behavior.

This simulation demonstrates practical endpoint detection and enterprise SOC monitoring capability for suspicious process analysis.

Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
221	16	3304	16884	0.22	5488	0	aesm_service
165	9	1856	13056	0.03	3772	0	agent_ovpnconnect
213	16	26496	24556	1.66	10936	0	agent-manager
222	13	4212	15776	0.67	8944	0	AggregatorHost
225	17	57308	39076	0.42	13432	1	ai
225	16	31852	21140	0.78	15544	1	ai
379	21	44920	32132	0.47	15988	1	aimgr
392	21	12356	45468	0.53	4804	1	AppActions
234	20	3548	14492	2.27	4924	0	AppleMobileDeviceService
406	21	13744	46552	0.39	11308	1	ApplicationFrameHost
812	14	8496	19912	28.56	4324	0	audiodg
332	28	18860	2460	0.30	8132	1	backgroundTaskHost
669	26	33484	48456	1.67	13020	1	backgroundTaskHost
330	29	18976	1888	0.39	15728	1	backgroundTaskHost
315	16	4712	26644	0.13	20904	1	backgroundTaskHost
446	50	492660	540440	642.61	1772	1	chrome
187	18	18876	33824	0.05	6000	1	chrome
314	39	214428	274744	28.06	6100	1	chrome
410	45	521328	605308	826.84	7108	1	chrome
322	34	78880	127616	12.97	11448	1	chrome
1324	48	283612	245612	139.02	15540	1	chrome
437	34	31944	63476	48.84	15628	1	chrome
177	12	18892	20564	0.64	15892	1	chrome
263	27	70716	96984	189.56	16244	1	chrome
1978	73	164908	251956	144.64	16292	1	chrome
223	10	2484	9448	0.13	16316	1	chrome
469	26	42376	75724	1.70	19004	1	chrome
268	17	10124	24624	20.45	19720	1	chrome
646	25	31340	42868	1.16	20044	1	chrome
262	25	25472	46136	1.44	20456	1	chrome
334	47	313256	361244	56.45	22432	1	chrome
218	18	26208	18912	0.39	5136	0	CmWebAdmin
388	22	10552	51216	13.67	5560	0	CodeMeter
229	17	7108	23432	1.41	18664	1	CodeMeterCC
283	15	11408	31308	2.63	5712	1	conhost
410	18	8328	24776	2.78	21764	0	wininit

Figure 14: Suspicious Process Execution Activity on Windows Host Machine

7.6 Network Traffic Analysis using Wireshark

Network traffic analysis is an essential SOC investigation capability used to identify suspicious communication patterns, abnormal connection attempts, and attacker network behavior.

During attack simulation activities, Wireshark was used to inspect packet-level communication generated between the monitored Windows host machine and the Kali Linux attacker environment.

The analysis provided visibility into protocol usage, connection attempts, packet flow behavior, and communication metadata useful for threat investigation.

Packet inspection strengthens enterprise monitoring by enabling analysts to validate suspicious activity beyond endpoint logs and SIEM alerts.

This simulation demonstrates practical network monitoring and packet analysis capability within the SOC lab environment.

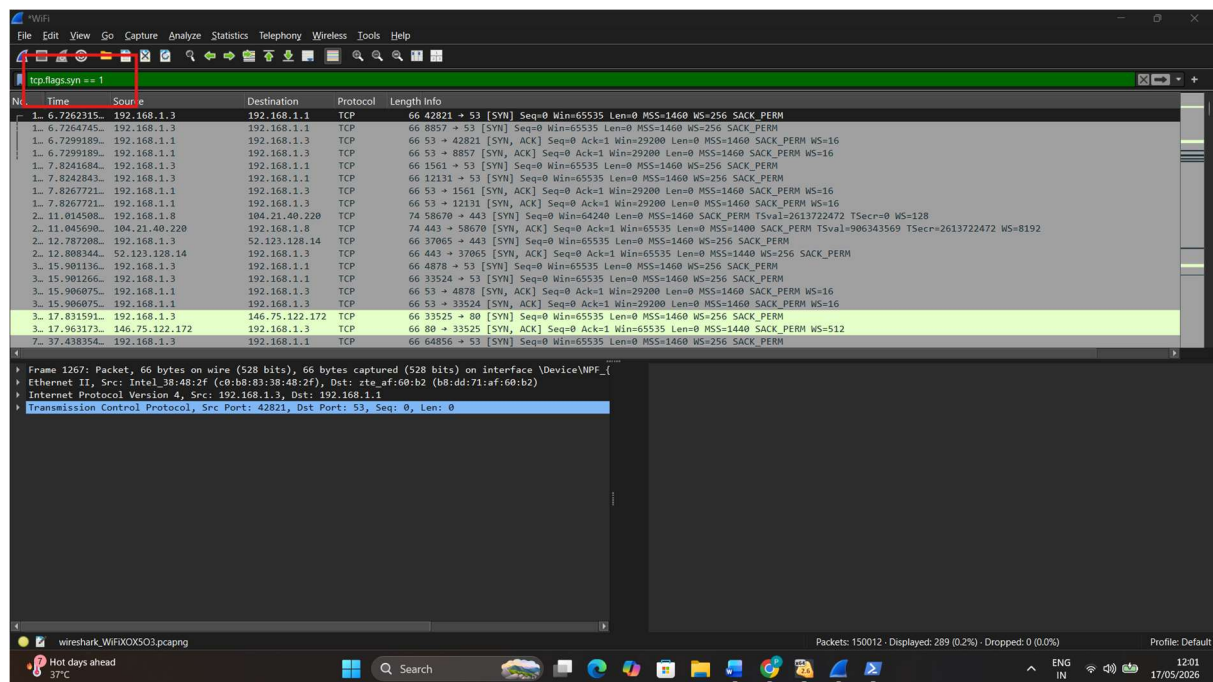


Figure 15: Packet-Level Network Traffic Analysis using Wireshark

7.7 Wazuh Alert Investigation and Correlation

Alert investigation and event correlation are critical SOC analyst responsibilities for identifying suspicious behavior, validating threat activity, and prioritizing incident response actions.

Wazuh was utilized to monitor endpoint security events, generate host-based alerts, and provide additional visibility into suspicious activity occurring on the monitored Windows host machine.

The platform supports practical alert triage workflows by highlighting security anomalies, authentication events, endpoint behavior indicators, and host-level threat intelligence.

Correlating Wazuh alerts with Splunk SIEM telemetry improves detection confidence and strengthens enterprise threat investigation workflows.

This implementation demonstrates practical multi-tool SOC alert investigation capability aligned with real-world blue team monitoring practices.

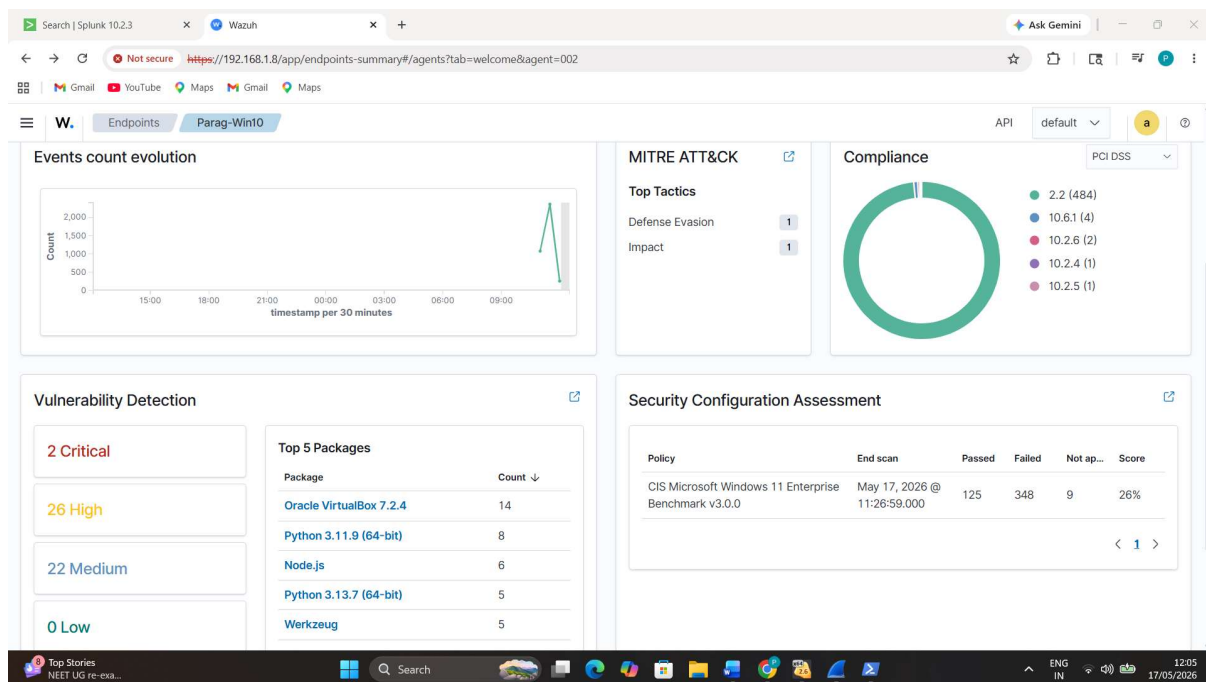


Figure 16: Wazuh Endpoint Security Investigation and Alert Correlation Dashboard

7.8 SOC Incident Response Workflow

Incident response is a critical SOC function that involves identifying suspicious activity, validating security alerts, investigating impacted systems, and initiating containment or remediation actions.

Within this SOC monitoring lab, the incident response workflow begins with security event detection through Splunk Enterprise, Wazuh endpoint monitoring, Sysmon telemetry, and network traffic analysis.

Detected suspicious activities such as reconnaissance scanning, authentication abuse, suspicious PowerShell execution, and unauthorized process activity are investigated through centralized event analysis.

The practical workflow includes alert identification, event correlation, endpoint review, network analysis, evidence validation, and security investigation decision-making.

This implementation demonstrates enterprise-style SOC incident response capability aligned with real-world blue team investigation workflows.

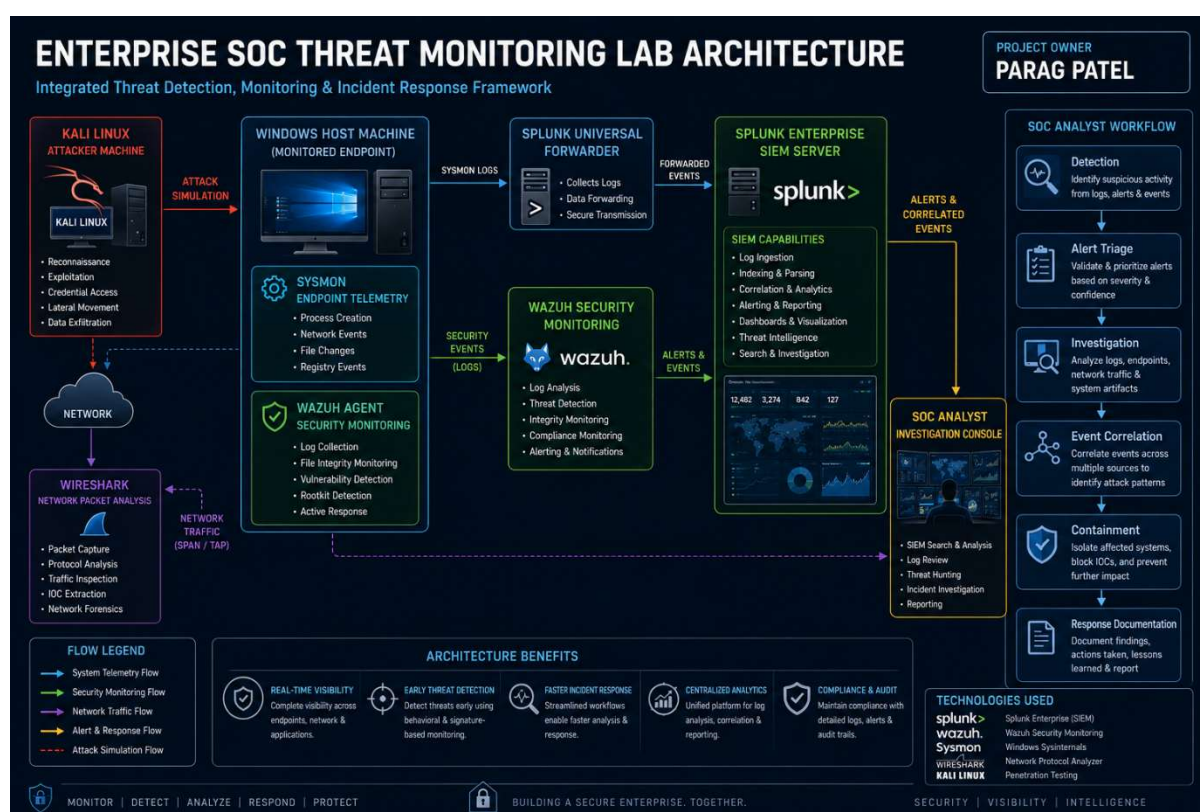


Figure 17: Enterprise SOC Incident Response Workflow

8. Results and Findings

The enterprise SOC monitoring implementation successfully demonstrated practical threat detection, endpoint visibility, network monitoring, and incident investigation capabilities across multiple cybersecurity monitoring layers.

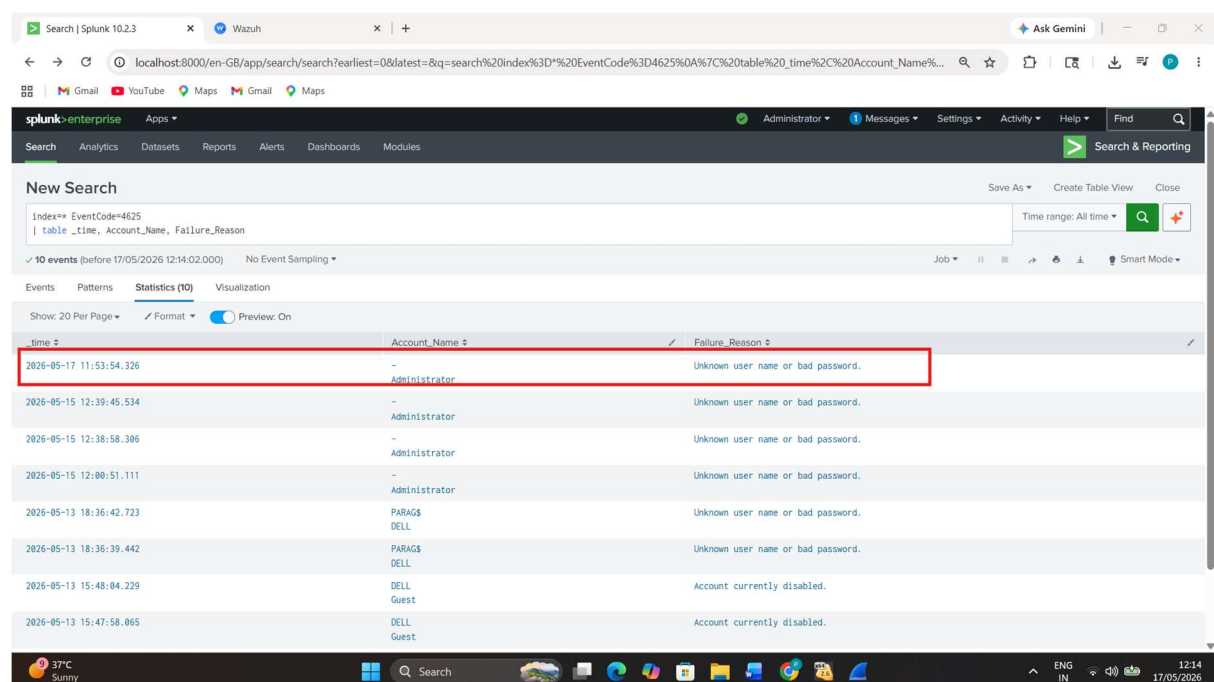
Controlled attack simulation scenarios generated realistic security telemetry, enabling validation of detection workflows across Splunk Enterprise, Wazuh, Sysmon, and Wireshark.

The monitoring environment successfully captured authentication abuse events, reconnaissance activity, suspicious PowerShell execution, unauthorized process behavior, endpoint alerts, and network communication visibility.

Centralized SIEM monitoring through Splunk Enterprise provided effective search analysis, event visibility, and detection engineering support.

Wazuh strengthened endpoint monitoring through alert generation, security visibility, and host-based investigation capability.

The overall implementation successfully replicates practical enterprise SOC analyst workflows within a controlled cybersecurity lab environment.



The screenshot displays the Splunk Enterprise Search interface. The search bar contains the query: `index=** EventCode=4625 | table _time, Account_Name, Failure_Reason`. The results table shows 10 events. The first event is highlighted with a red box.

time	Account_Name	Failure_Reason
2026-05-17 11:53:54.326	Administrator	Unknown user name or bad password.
2026-05-15 12:39:45.534	Administrator	Unknown user name or bad password.
2026-05-15 12:38:58.306	Administrator	Unknown user name or bad password.
2026-05-15 12:00:51.111	Administrator	Unknown user name or bad password.
2026-05-13 18:36:42.723	PARAGS DELL	Unknown user name or bad password.
2026-05-13 18:36:39.442	PARAGS DELL	Unknown user name or bad password.
2026-05-13 15:48:04.229	DELL Guest	Account currently disabled.
2026-05-13 15:47:58.065	DELL Guest	Account currently disabled.

Figure 18: Splunk Detection of Failed Authentication Attempts using EventCode 4625

9. MITRE ATT&CK Mapping

The simulated attack scenarios performed within the SOC monitoring lab align with several well-known adversary techniques documented within the MITRE ATT&CK framework.

Mapping simulated activities to MITRE ATT&CK improves threat understanding, strengthens detection engineering, and aligns practical SOC monitoring workflows with industry-recognized threat intelligence methodologies.

The following mappings were identified:

Simulated Activity	MITRE ATT&CK Technique
Nmap Reconnaissance Scan	T1046 – Network Service Scanning
Brute Force Authentication Abuse	T1110 – Brute Force
Suspicious PowerShell Activity	T1059.001 – PowerShell
Unauthorized Process Execution	T1059 – Command and Scripting Interpreter
Network Traffic Investigation	T1049 – System Network Connections Discovery

This mapping demonstrates that the practical SOC monitoring environment aligns with real-world attacker behavior patterns commonly encountered by enterprise security operations teams.

10. Conclusion and Future Enhancements

This project successfully designed and implemented a practical enterprise-style Security Operations Center (SOC) monitoring environment using industry-relevant cybersecurity tools and defensive monitoring technologies.

The implementation combined Splunk Enterprise, Wazuh, Sysmon, Wireshark, Kali Linux, and Windows endpoint monitoring to create a realistic multi-layer threat detection and incident investigation lab.

Practical attack simulations including reconnaissance scanning, brute force authentication abuse, suspicious PowerShell activity, unauthorized process monitoring, network traffic inspection, and alert investigation successfully validated the effectiveness of the monitoring architecture.

The project strengthened practical skills in SIEM monitoring, endpoint detection, log analysis, alert triage, network investigation, detection engineering, and enterprise SOC incident response workflows.

This implementation closely reflects practical blue team security monitoring responsibilities expected in real-world SOC analyst environments.

Future Enhancements

Future improvements may include:

- Custom Splunk detection dashboards
- Automated alerting rules and correlation searches
- Advanced Wazuh threat hunting use cases
- Integration with threat intelligence feeds
- Malware analysis telemetry integration
- Expanded MITRE ATT&CK detection coverage

This project establishes a strong practical foundation for enterprise SOC analyst and blue team cybersecurity roles.

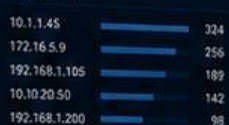
Security Posture Overview



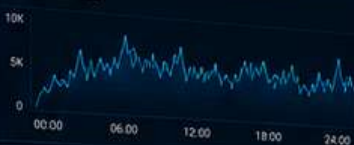
Top Event Categories



Top Sources



Events Over Time



Threat Map



wazuh.

Security Overview



Top Alert Levels



Recent Alerts

Time	Rule Description	Level
10:21:45	Brute Force Attack Detected	Critical
10:21:10	Malware Detected	Critical
10:20:32	Suspicious Activity	High
10:19:18	Privilege Escalation	High
10:18:55	Possible Data Exfiltration	Medium

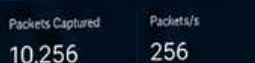
Sysmon Endpoint Telemetry



Sysmon Events Over Time



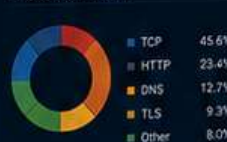
WIRESHARK Network Packet Analysis



Top Conversations

Source	Destination	Packets	Bytes
192.168.1.105	10.10.20.50	2,451	330 MB
10.1.1.45	172.16.5.9	1,855	210 MB
192.168.1.200	8.8.8.8	1,245	180 MB
172.16.5.9	192.168.1.105	980	120 MB
10.10.20.50	192.168.1.200	750	98 MB

Protocol Distribution



Threat Intelligence

Indicators of Compromise



Top Threat Feeds

Feed	Count
AbuseIPDB	32
AlienVault OTX	28
ThreatFox	16
VirusTotal	8

Threat Categories



Authentication Attack Detection

Brute Force Attempts



Failed Logins



Top Targeted Accounts

Account	Count
administrator	214
admin	155
test	98
guest	64
oracle	38



Incident Response Overview

Open Incidents



In Progress



Resolved



Closed



Mean Time to Respond



Mean Time to Resolve



Recent Incidents

ID	Incident Name	Severity	Status	Assigned To	Last Update
INC-1001	Brute Force - SSH	Critical	In Progress	SOC Analyst	10:24 AM
INC-1002	Malware Detected - Endpoint	Critical	In Progress	SOC Analyst	10:18 AM
INC-1003	Suspicious PowerShell Activity	High	Investigating	SOC Analyst	10:15 AM
INC-1004	Possible Data Exfiltration	Medium	Investigating	SOC Analyst	10:10 AM

Incidents by Severity



CONCLUSION

Enterprise SOC Threat Detection & Incident Response Lab

This project successfully demonstrates the implementation of an enterprise-grade SOC environment for real-time threat detection, monitoring, and incident response using industry-leading tools and technologies.

KEY ACHIEVEMENTS



Real-Time Threat Detection

Successfully implemented continuous monitoring to detect and identify threats across endpoints, network, and systems in real time.



Centralized SIEM Monitoring

Integrated Splunk Enterprise SIEM for centralized log collection, analysis, alerting, and visualization for improved security visibility.



Endpoint Security Visibility

Leveraged Wazuh and Sysmon telemetry to achieve deep visibility into endpoint activities, vulnerabilities, and system events.



Alert Investigation and Correlation

Correlated events from multiple data sources to identify attack patterns, reduce false positives, and prioritize critical threats.



Incident Response Workflow

Established an effective incident response process including detection, triage, investigation, containment, and recovery.



Network Traffic Analysis

Utilized Wireshark for deep packet inspection and network behavioral analysis to detect anomalies and malicious activity.



Practical Blue Team Security Operations

Hands-on implementation of enterprise-grade tools and techniques to strengthen blue team capabilities and improve overall security posture.



Project by Parag Patel

Cybersecurity SOC Analyst Portfolio Project

